

Low-Moral Actions by Malicious Anonymous Operators of Avatar Robots

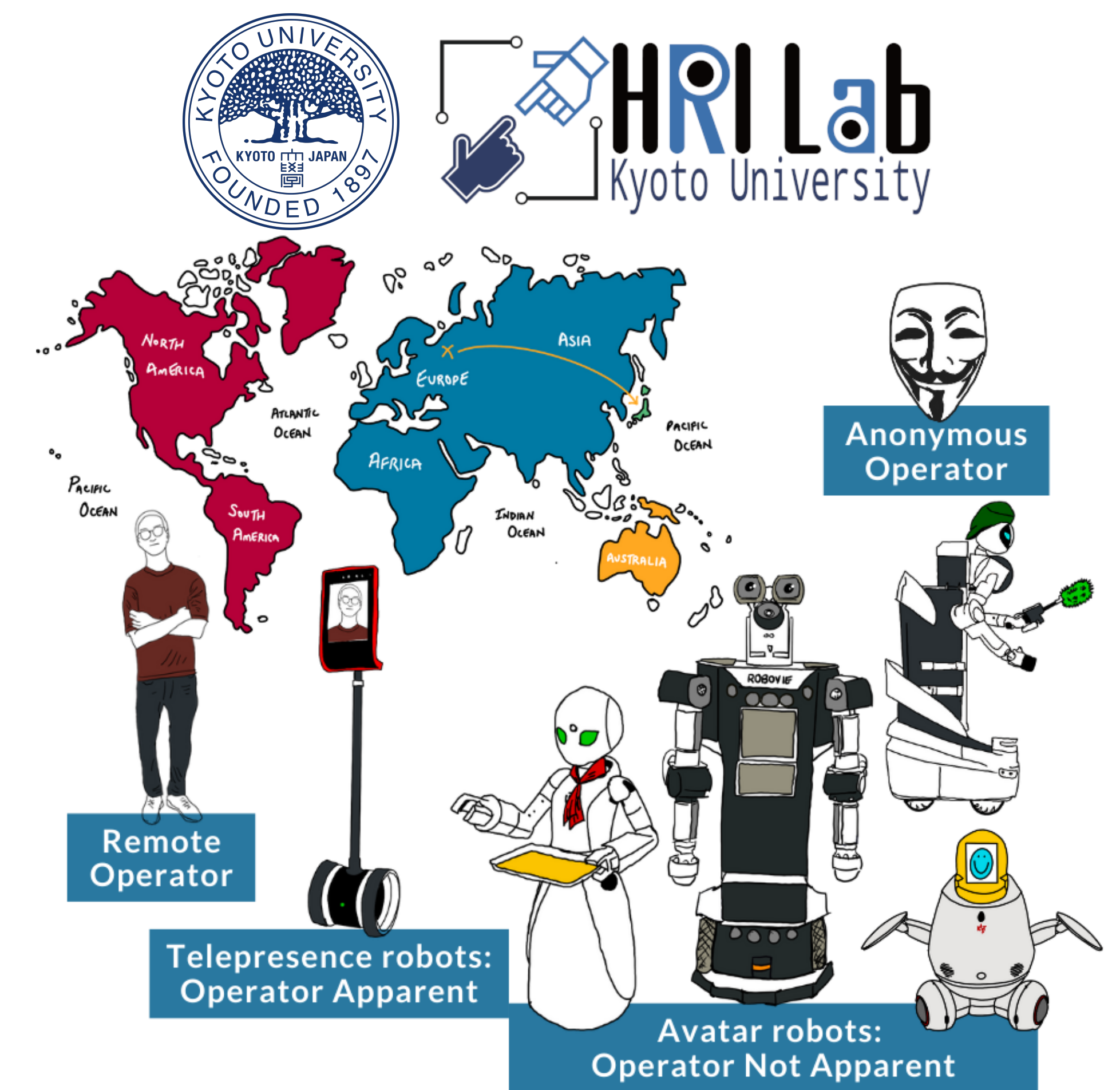
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Main Objectives

1. Examine the potential for misuse of avatar robots when controlled by anonymous teleoperators.
2. Conduct hazard identification workshops to construct a list of low-moral actions.
3. Categorize identified low-moral actions into major categories and subcategories.
4. Suggest prevention strategies by reviewing literature from relevant fields on how to mitigate these low-moral actions.

Background

- Avatar robots allow social interactions with remote people and environments.
- Malicious users can use anonymity to do low-moral actions.
- How can we prevent these intentionally malicious acts?

Approach - Hazard Identification Workshops

- Conducted workshops with potential regular users of avatar robots.
- Participants gained experience to assist brainstorming through:
 - Controlling avatar
 - Sharing space with avatar
 - Controlling avatar in simulation
- 320 ideas collected; affinity diagrams used to consolidate ideas.

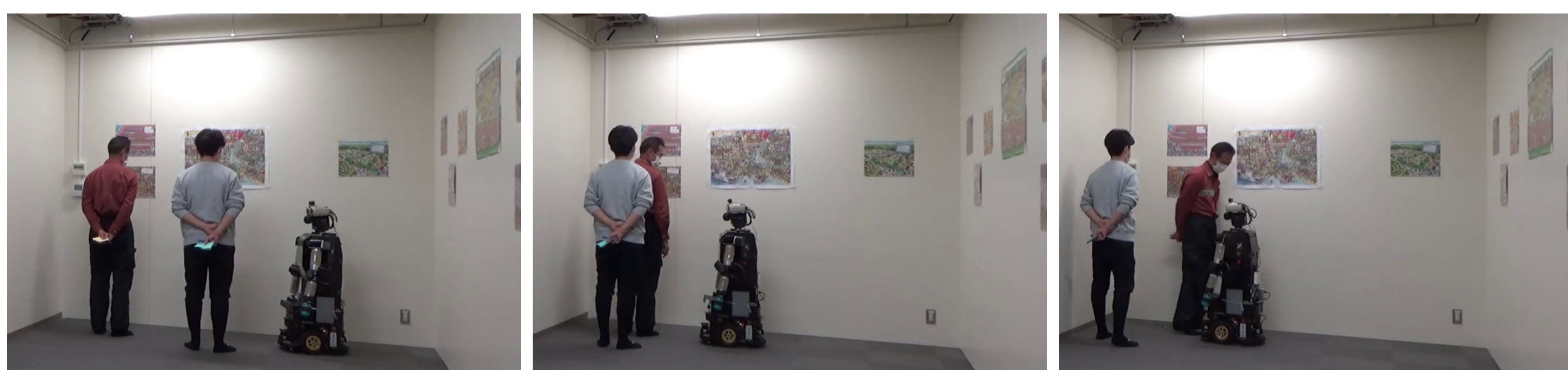


Figure 1: Left to right: Avatar robot forces participant to move from their position.



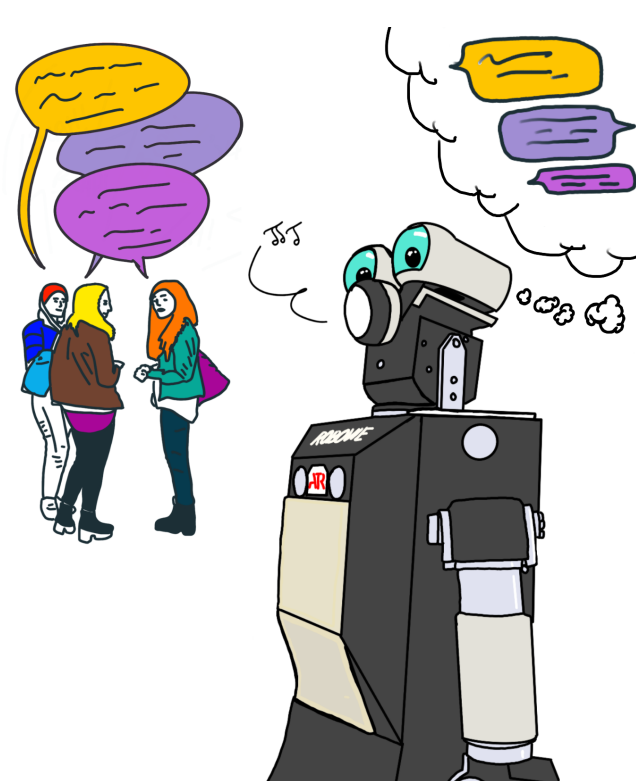
Figure 2: Malicious actions in various simulated environments.

Results - Four Distinct Categories Identified

1. Violate Privacy and Security

- Actions breaching privacy of individuals or the security of locations.
 - “Recording a person’s face without permission”
 - “Following you to the bathroom (in the park)”

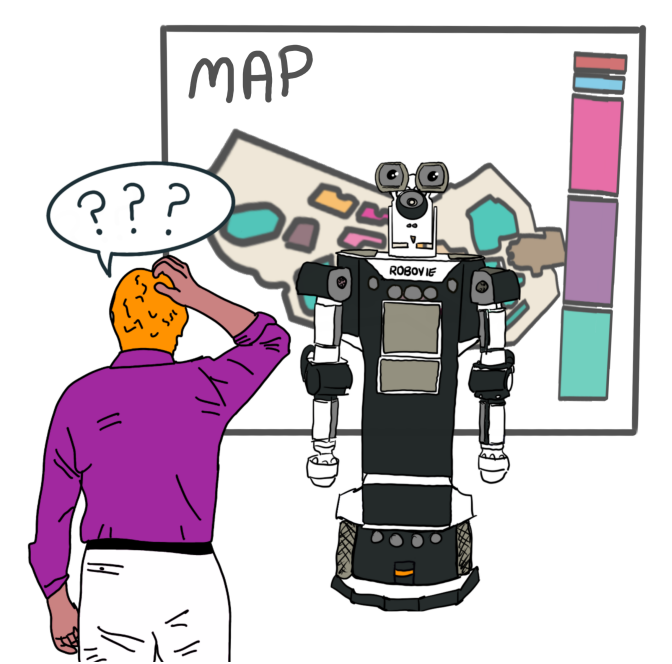
Subcategories: Steal Information, Photograph without Consent, Eavesdrop, Stalk, & Enter Forbidden Space.



2. Inhibit

- Using the robot body to prevent from doing something
- Examples: getting a magazine from a stand, water from a water fountain; from interacting with an ATM, emergency defibrillators; or from sitting on a chair.

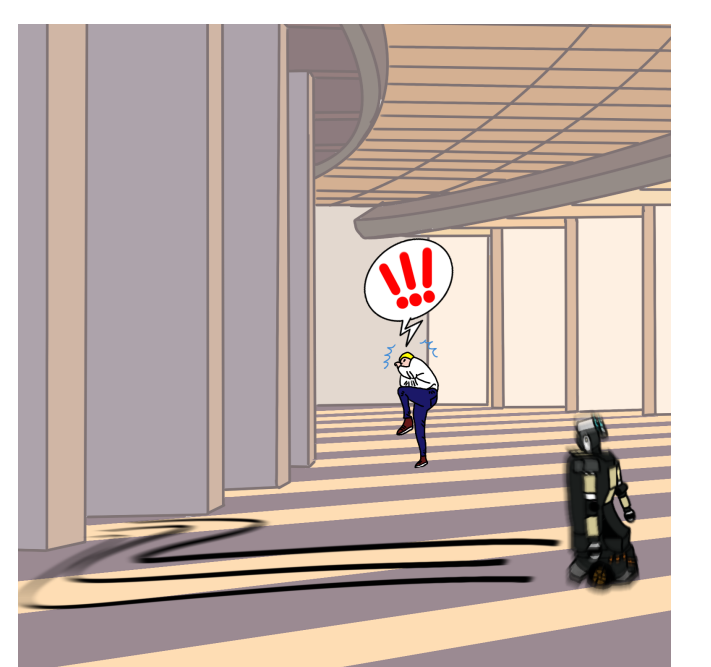
Subcategories: Inhibit Access to Resources & Inhibit Movement.



3. Annoy

- Using the robot to annoy, distract, or scare people.
 - “Maneuvering like a gasping car”
 - “Run into a pet’s lead” (break up a group containing a pet and a human)

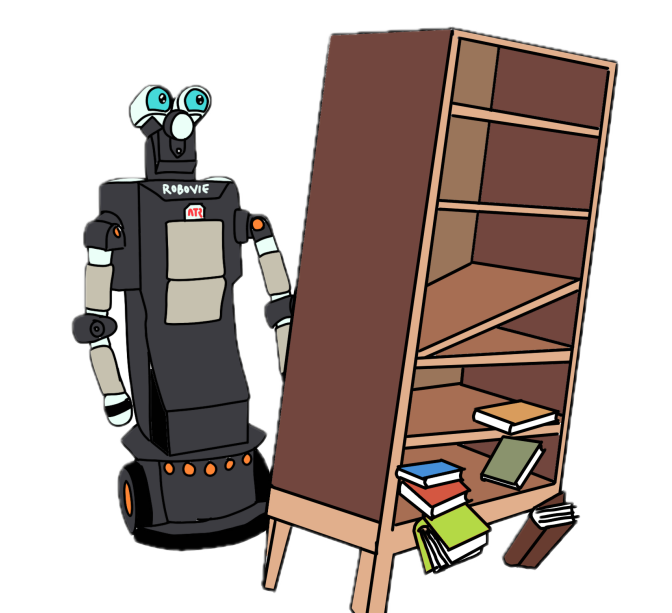
Subcategories: Unnatural Movement, Invade Personal Space, Chase, Mislead, & Use Sound.



4. Destroy or Hurt

- Destruction of property and harming people or animals.
 - “Attempt to destroy the robot itself”
 - “Enables the use of weapons at a distance”

Subcategories: Destroy Items, Hurt Living Things, & Mass Violence



Prevention Strategies

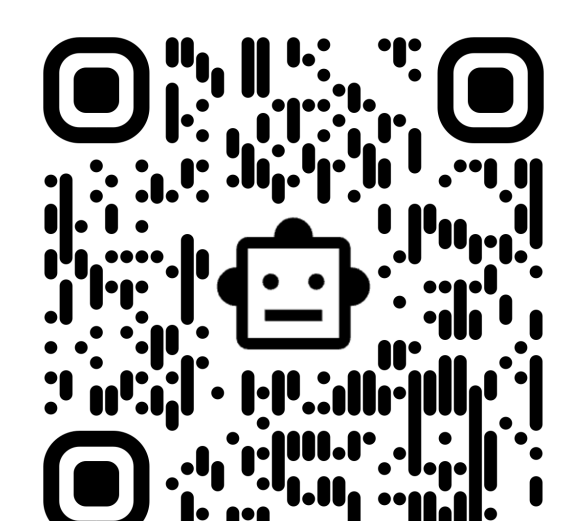
- **Filtering** information to operator so they do not perceive entire environment
- **Restricting** control of operator to only high level commands
- **Monitoring** operator actions to identify low-moral behaviour
- **Mitigating Impact** by providing accountability and transparency

Issues with Prevention - Some Questions Raised

- Restricting agency of operator detracts from user experience and dehumanizes operator. What about actions that are moral in context? What is the balance?
- Observing every action by operator gives sense of surveillance. Punishing the non-malicious? Can we judge or measure maliciousness?
- Abuse from bystanders is well-known issue. How to distinguish between operator vs bystander being malicious?

Acknowledgments

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¹Work was conducted while first author was a Master’s student at the Human-Robot Interaction Lab at Kyoto University, Japan.